

Do Sportsbooks Overreact to Recent Performance? Evidence from Blowout Wins in the NFL

Abstract

This study examines whether NFL betting markets systematically overreact to extreme recent performance, specifically large-margin (“blowout”) victories. Drawing on regular-season data from 2015–2025 obtained via nflfastR, I test whether teams coming off blowout wins are subsequently overpriced, as measured by their likelihood of covering the point spread and by spread error magnitude. Logistic regression models, linear regression models, interaction tests for high-attention games, and betting strategy backtests are employed. While recent margins of victory strongly predict subsequent point spreads, I find no statistically or economically significant evidence that teams coming off blowout wins are systematically overpriced. Betting strategies that fade blowout teams do not generate abnormal returns. These findings suggest that NFL betting markets efficiently incorporate extreme recent performance without systematic overreaction, even in highly salient contexts.

1. Introduction

Sports betting markets provide one of the cleanest empirical settings for testing market efficiency. Prices are transparent, outcomes resolve quickly, and payoffs are clearly defined. In theory, competitive betting markets with sophisticated participants should incorporate all publicly available information into prices, leaving no systematic arbitrage opportunities.

However, behavioral economics challenges the assumption that agents process information fully rationally. Individuals frequently overweight recent, vivid, and salient information when forming expectations. In sports contexts, blowout wins, defined as dominant performances with large margins of victory, receive disproportionate media coverage and fan attention. These performances may alter perceptions of team quality beyond what fundamentals justify.

The National Football League (NFL) provides an ideal setting to test this hypothesis. The league’s short regular season amplifies the informational weight of each game. Media narratives rapidly form around dominant performances, and betting volumes are among the highest in global sports markets.

This study asks:

Are NFL teams coming off large-margin (blowout) victories systematically overpriced in their subsequent games?

Overpricing is defined as a reduced likelihood of covering the point spread relative to market expectations. Secondary questions examine whether certain teams are blind spots for sportsbooks and whether sportsbooks become sharper over the course of a season.

Understanding whether sportsbooks overreact to extreme recent performance has practical and theoretical significance. If recency bias affects pricing, this would challenge strong-form efficiency in high-liquidity betting markets. Conversely, a null result reinforces the view that competitive market forces quickly eliminate behavioral distortions.

For this project, blowouts were defined as a 14+ point differential and an extreme previous spread error in the top or bottom 10% of the season. The lower threshold was the home team losing by 16 more points than the spread predicted, and the upper threshold was the home team winning by more than 16.5 points than the spread predicted.

2. Literature Review

2.1 Market Efficiency in Sports Betting

Sports betting markets have long been used to test weak-form efficiency (Sauer, 1998). In efficient markets, point spreads should serve as unbiased predictors of game outcomes. Gandar et al. (1988) find little evidence of systematic bias in NFL point spreads. Subsequent research confirms that closing lines, in particular, incorporate available information effectively.

However, sportsbooks do not purely forecast outcomes. Levitt (2004) demonstrates that sportsbooks may shade lines to balance bettor demand, suggesting that prices may reflect both probability assessments and risk management considerations.

The presence of behavioral bettors creates potential for localized inefficiencies even within generally efficient markets.

2.2 Behavioral Overreaction and Recency Bias

Behavioral finance models (Barberis et al., 1998) predict that individuals overweight recent signals, leading to short-run overreaction followed by correction. This mechanism has been documented in equity markets, where recent earnings surprises can temporarily distort prices.

Sports betting markets may be particularly susceptible to recency bias due to:

- Emotional engagement
- Media amplification
- Simplified narratives (e.g., “the team is hot”)
- Short evaluation windows

Blowout wins are extreme realizations of performance and, therefore, prime candidates for overreaction.

2.3 Empirical Evidence in Betting Markets

Empirical work documents several betting anomalies, including:

- Favorite–longshot bias (Cain et al., 2000)
- Public team bias
- Momentum mispricing in streaks (Avery & Chevalier, 1999)

However, much of this literature relies on binary win/loss streak indicators. Margin of victory provides richer continuous information and may generate stronger psychological responses.

Despite this, relatively few studies isolate the effect of extreme recent margins in NFL betting markets. This study addresses that gap.

3. Data

3.1 Sample

Data include all NFL regular-season games from 2015–2025, obtained via nflfastR. After removing observations with missing spread data, the final sample contains 2,860 games.

3.2 Key Variables

- **spread_line**: Closing point spread (home perspective)
- **home_cover**: Indicator equal to 1 if the home team covers (often the response variable)
- **home_rest, away_rest, div_game**: Controls included in all models
- **Home_blowout_win_prev**: Home team won previous game by ≥ 14 points, and the game was a top or bottom 10% spread error
- **Away_blowout_win_prev**: Away team won previous game by ≥ 14 points, and the game was a top or bottom 10% spread error
- **Home_blowout_loss_prev**: Home lost previous game by ≥ 14 points, and the game was a top or bottom 10% spread error
- **Away_blowout_loss_prev**: Away team lost previous game by ≥ 14 points, and the game was a top or bottom 10% spread error

4. Methodology

Initially, I completed analysis with a blowout defined as a 14+ point game, then again with a blowout being a top or bottom 10% spread error. Both of these strategies indicated that sportsbooks are efficient and are not susceptible to biases. After discussion with Dr. Gong, I decided to combine the definitions in an attempt to provide a more accurate definition, to narrow the scope, and to enhance my findings.

4.1 Testing Overreaction via Cover Probability

Investigating whether sportsbooks overreact to blowout wins, causing home teams to cover less frequently than expected.

$$\text{logit}(P(\text{HomeCover} = 1)) = \beta_0 + \beta_1 \cdot \text{HomeBlowoutWinPrev} + \beta_2 \cdot \text{HomeBlowoutLossPrev} + \beta_3 \cdot \text{AwayBlowoutWinPrev} + \beta_4 \cdot \text{AwayBlowoutLossPrev} + \beta_5 \cdot \text{SpreadLine} + \beta_6 \cdot \text{Controls} + \delta_i$$

Where:

- HomeCover_i = indicator that the home team covers the spread in game i
- HomeBlowoutWin_{i-1} = indicator the home team won its previous game within the blowout definition
- HomeBlowoutLoss_{i-1} = indicator the home team lost its previous game within the blowout definition
- AwayBlowoutWin_{i-1} = indicator the away team won its previous game within the blowout definition
- AwayBlowoutLoss_{i-1} = indicator the away team lost its previous game within the blowout definition
- SpreadLine_i = closing point spread for game i
- Controls:
 - HomeRest_i, AwayRest_i = days of rest entering the game
 - DivisionGame_i = indicator for divisional matchup
 - δ_s = season fixed effects

4.2 Spread Error Model

Investigating whether there is a tendency for the spread to be incorrect following blowout games.

$$\text{SpreadError} = \beta_0 + \beta_1 \cdot \text{HomeBlowoutWinPrev} + \beta_2 \cdot \text{HomeBlowoutLossPrev} + \beta_3 \cdot \text{AwayBlowoutWinPrev} + \beta_4 \cdot \text{AwayBlowoutLossPrev} + \beta_5 \cdot \text{SpreadLine} + \beta_6 \cdot \text{Controls} + \delta_i + \epsilon$$

Here:

- SpreadError = actual margin – expected margin from the spread
- δ_i = season fixed effects

- ε = error term

4.3 Recency Bias in Spread Formation

Investigating how the spread line changes following blowout games and if it moves differently from expected.

$$\text{SpreadLine} = \beta_0 + \beta_1 \cdot \text{HomeBlowoutWinPrev} + \beta_2 \cdot \text{HomeBlowoutLossPrev} + \beta_3 \cdot \text{AwayBlowoutWinPrev} + \beta_4 \cdot \text{AwayBlowoutLossPrev} + \beta_5 \cdot \text{Controls} + \delta_{it} + \varepsilon$$

Where:

- SpreadLine = closing betting spread for the game

4.4 Season Progression Consideration

Investigating whether the week of the season impacts the likelihood of the home team covering.

$$\begin{aligned} \text{logit}(P(\text{HomeCover} = 1)) = & \beta_0 + \beta_1 \cdot \text{HomeBlowoutWinPrev} + \beta_2 \cdot \text{HomeBlowoutLossPrev} + \\ & \beta_3 \cdot \text{AwayBlowoutWinPrev} + \beta_4 \cdot \text{AwayBlowoutLossPrev} + \beta_5 \cdot \text{Week} + \\ & \beta_6 \cdot (\text{HomeBlowoutWinPrev} \times \text{Week}) + \beta_7 \cdot (\text{HomeBlowoutLossPrev} \times \text{Week}) + \\ & \beta_8 \cdot (\text{AwayBlowoutWinPrev} \times \text{Week}) + \beta_9 \cdot (\text{AwayBlowoutLossPrev} \times \text{Week}) + \beta_{10} \cdot \text{SpreadLine} \\ & + \beta_{11} \cdot \text{Controls} \end{aligned}$$

Where:

- Week = week of the NFL season
- HomeBlowoutWinPrev $\times$ Week = interaction allowing the effect of a home blowout win to vary over the season
- HomeBlowoutLossPrev $\times$ Week = interaction allowing the effect of a home blowout loss to vary over the season
- AwayBlowoutWinPrev $\times$ Week = interaction allowing the effect of an away blowout win to vary over the season
- AwayBlowoutLossPrev $\times$ Week = interaction allowing the effect of an away blowout loss to vary over the season

4.4 Team Random Effects

Investigating whether certain teams tend to cover the spread at different rates following blowout games.

$$\begin{aligned} \text{logit}(P(\text{HomeCover} = 1)) = & \beta_0 + \beta_1 \cdot \text{HomeBlowoutWinPrev} + \beta_2 \cdot \text{HomeBlowoutLossPrev} + \\ & \beta_3 \cdot \text{AwayBlowoutWinPrev} + \beta_4 \cdot \text{AwayBlowoutLossPrev} + \beta_5 \cdot \text{SpreadLine} + \beta_6 \cdot \text{HomeRest} + \\ & \beta_7 \cdot \text{AwayRest} + \beta_8 \cdot \text{DivGame} + \delta_{it} + u_{\text{home}} + v_{\text{away}} \end{aligned}$$

Where:

- u_{home} = home team random effects, ($u_{\text{home}} \sim \text{mathcal{N}}(0, \sigma^2_{\text{home}})$)
- v_{away} = away team random effects, ($v_{\text{away}} \sim \text{mathcal{N}}(0, \sigma^2_{\text{away}})$)
- δ_{season} = season fixed effects

5. Results

5.1 Blowout and Cover Probability

NFL betting markets appear efficient with respect to extreme prior performance. While sportsbooks clearly incorporate information about recent games into point spreads, these adjustments do not lead to systematic overpricing or underpricing. Teams coming off true blowouts are no more or less likely to cover the spread in their next game. P-values for our four variables indicate that these results may have happened by random chance and that blowout games do not affect home cover probability.

Variable	Interpretation	p-value
home_blowout_win_prev	Home team coming off a blowout win	0.824
home_blowout_loss_prev	Home team coming off a blowout loss	0.823
away_blowout_win_prev	Away team coming off a blowout win	0.852
away_blowout_loss_prev	The away team is coming off a blowout loss	0.577

If sportsbooks or bettors overreacted to blowouts, we would expect:

- Teams coming off blowout wins to be overpriced → less likely to cover
- Teams coming off blowout losses to be underpriced → more likely to cover

The results show neither pattern.

5.2 Spread Error

While p-values are lower (between .12 and .67), there is still no convincing evidence that teams coming off blowout wins or losses systematically outperform or underperform the betting spread in their subsequent game. The adjusted R-squared was only 0.001244, confirming no predictive accuracy. The magnitude of spread errors appears largely random with respect to extreme prior performance. These results reinforce the conclusion that NFL betting markets efficiently incorporate recent performance information without creating predictable pricing errors.

5.3 Recency in Spread Setting

Variable	Estimate	Interpretation
home_blowout_win_prev	+2.48	Home team becomes ~2.5 points more favored
home_blowout_loss_prev	-2.62	Home team becomes ~2.6 points less favored
away_blowout_win_prev	-1.46	Away team blowout win makes home team ~1.5 points less favored
away_blowout_loss_prev	+1.86	Away team blowout loss makes home team ~1.9 points more favored

This model shows that blowouts do significantly shift the spread for next week's game. However, these shifts are not excessive. Sportsbooks react strongly to extreme performance, but the adjustments appear to be appropriately calibrated rather than excessive. In other words, markets update spreads after blowouts, but the adjustments do not lead to predictable mispricing. For this model, the Adjusted R-squared was 0.05112, indicating that there is very little predictive accuracy across all blowouts.

5.4 Season Progression

There is some weak evidence that away teams coming off a blowout win perform slightly better against the spread early in the season, but the effect declines with later weeks. It appears that the market corrects the minor advantage quickly. Overall, previous blowouts provide little predictive value for covering the spread and should not reliably inform strategy.

5.5 Team Random Effects

This mixed logit model finds little evidence that the included variables explain whether a team covers the spread once team-level effects are taken into account. None of the fixed effects are statistically significant, suggesting that these factors do not provide predictive power for covering beyond what is already embedded in the spread. The random effects reveal an important insight. Over the course of the dataset, the variance for home team effects is essentially zero. In contrast, the variance for home teams is small but non-zero, indicating minimal heterogeneity in teams' ability to systematically cover the spread, especially at home. This supports the idea that betting markets are largely efficient and that team-specific tendencies to beat the spread are weak or nonexistent. However, the model also exhibits a minor convergence issue, so results should be interpreted cautiously, though the overall conclusion of a limited signal is unlikely to change.

6. Discussion

The results of this study suggest that NFL betting markets are largely efficient in incorporating information about extreme prior performance. Although blowout wins and losses represent highly salient events that attract significant media attention and fan discussion, the empirical results indicate that these outcomes do not create predictable opportunities for bettors in the following week's games.

The results across all specifications consistently point to one conclusion: NFL betting markets are highly efficient with respect to extreme prior performance, and there is no discernible edge to be gained from betting strategies based purely on historical blowout outcomes. While sportsbooks clearly adjust point spreads in response to recent blowouts, often by multiple points, these adjustments appear to be well-calibrated rather than excessive. This indicates that teams coming off blowout victories are not systematically overpriced, and teams coming off blowout defeats are not systematically underpriced.

If sportsbooks or bettors overreacted to extreme prior results, one would expect a clear directional pattern in the data. Specifically, teams coming off blowout wins would be expected to cover the spread less frequently if sportsbooks inflated their lines due to public enthusiasm. Conversely, teams coming off blowout losses would be expected to cover more often if sportsbooks discounted them too heavily. The results show no evidence of either pattern, suggesting that sportsbooks and betting markets effectively account for extreme prior outcomes when setting lines.

Although blowouts do not predict cover outcomes, they do have a clear effect on how sportsbooks set subsequent point spreads. The spread-setting model shows that sportsbooks respond meaningfully to extreme results, with scores being incorporated into prices, as teams coming off blowout wins or losses experience meaningful shifts in their subsequent spreads. However, the absence of any corresponding effect on cover probability or spread error indicates that these adjustments are appropriately priced by the market and are generally well-calibrated. The absence of predictable spread errors or cover probability effects implies that sportsbooks adjust spreads in a way that appropriately reflects new information rather than exaggerating it. In other words, betting markets react to blowouts, but do not appear to overreact. If sportsbooks or bettors systematically overreacted to blowouts, we would expect to observe predictable biases, such as favorites coming off blowout wins underperforming or teams coming off blowout losses outperforming. The data provide no evidence of such patterns, suggesting that any initial reaction to extreme outcomes is efficiently corrected in the pricing process.

Furthermore, the mixed-effects model reinforces this conclusion by showing that team-level tendencies to outperform the spread are minimal to nonexistent. Once team-specific effects are accounted for, none of the explanatory variables retain predictive power. The near-zero variance in home team effects and the very small variance in away team effects indicate that there is little persistent, exploitable signal in how teams perform relative to the spread. This aligns with the broader notion of market efficiency, where publicly available information, in this case, recent game outcomes, is quickly and accurately incorporated into prices.

Although there is slight evidence of effects early in the season, these patterns are weak and inconsistent, further limiting their practical usefulness. Taken together, the findings suggest that historical performance, even in extreme cases like blowouts, does not provide a reliable basis for generating positive expected value in NFL betting markets. Any potential inefficiencies are either too small, too short-lived, or too noisy to exploit with simple rule-based strategies.

In practical terms, this implies that profitable betting strategies are unlikely to emerge from straightforward interpretations of past results. Instead, success in this domain likely requires more sophisticated approaches, such as identifying less obvious sources of mispricing. Overall, the evidence strongly supports the view that betting markets efficiently process and price recent information, leaving little room for systematic exploitation based on historical blowout data alone.

7. Limitations

A primary limitation of this study is that each game's betting spread is analyzed only in relation to the immediately preceding game's outcome. While this approach isolates the effect of a single prior blowout, it may overlook broader performance trends that develop over multiple weeks. Teams often experience injuries, lineup changes, or strategic adjustments that evolve throughout the season, and incorporating longer performance windows or additional explanatory variables could reveal patterns that are not captured in this framework. Variables such as

multi-week point differentials, injury reports, quarterback changes, or market betting volume may provide additional insight into how sportsbooks and bettors process information.

Another limitation is that the analysis focuses exclusively on traditional sportsbook betting markets, which are typically highly efficient due to the professional oddsmakers, large liquidity, and rapid information aggregation. As a result, mispricing opportunities may be rare and difficult to detect. The findings of this project would likely have greater relevance in prediction markets, such as those offered by Polymarket or Kalshi, where prices may be influenced more heavily by behavioral factors, sentiment, and retail trading behavior. Future research could extend this analysis by incorporating data from these platforms to examine whether extreme prior performances generate measurable sentiment-driven mispricing in markets where prices are determined more directly by trader beliefs rather than sportsbook risk management.

8. Conclusion

Overall, the findings suggest that blowout results alone should not serve as the foundation for betting strategies. While sportsbooks clearly respond to extreme performances by meaningfully adjusting point spreads, these adjustments appear to be well-calibrated rather than excessive, leaving little room for systematic exploitation. Across multiple model specifications, including cover probability, spread error, spread formation, season interactions, and team-level controls, the results consistently show that one-week blowout information does not translate into a predictive edge. Even when allowing for variation across teams or over the course of a season, any observed effects are weak, inconsistent, or economically negligible.

These findings contribute to a broader body of sports economics literature demonstrating that modern NFL betting markets are highly efficient. The combination of sophisticated oddsmaking, large market participation, and rapid information incorporation makes it difficult for simple, publicly observable signals, in this case, recent blowouts, to generate consistent profits. For bettors, this implies that strategies based solely on recent extreme outcomes are unlikely to succeed in the long run. More generally, the results highlight the challenge of identifying durable edges in mature betting markets, where most readily available information is already reflected in prices.